

Financial services: These blockchains can be used for applications such as cross-border payments, trade finance, and settlement systems, enabling faster, more cost-effective transactions while ensuring regulatory compliance.

Healthcare: Private blockchains enable secure sharing of patient data, medical records, and clinical trial results among healthcare providers, enhancing data integrity and interoperability while protecting patient privacy.

Identity management: They offer solutions for secure identity verification, authentication, and access control, enabling organisations to manage digital identities and credentials in a decentralised manner.

Public blockchain platforms

Public blockchain platforms are permissionless networks where anyone can participate and contribute to the network without restrictions. These platforms offer decentralisation, transparency, and censorship resistance, making them ideal for applications that prioritise openness and inclusivity. Key features of public blockchains include:

Permissionless access: Anyone can join and participate in the network without needing approval, fostering inclusivity and decentralisation.

Decentralised governance:

Governance structures are decentralised, with decision-making distributed among network participants through consensus mechanisms such as Proof of Work (PoW) or Proof of Stake (PoS).

Transparency: Public blockchains provide transparent access to transaction data and network activity, enabling anyone to verify transactions and monitor the integrity of the network.

Censorship resistance: These blockchains are resistant to censorship and tampering, as transactions are validated by a distributed network of nodes rather than centralised authorities.

Here are a few popular use cases of public blockchain platforms.

Cryptocurrencies: Public blockchains serve as the underlying infrastructure for cryptocurrencies such as Bitcoin and Ethereum, enabling peer-to-peer transactions and decentralised financial systems.

Decentralised finance (DeFi):

They support a wide range of decentralised financial applications, including lending, borrowing, trading, and asset management, without the need for intermediaries.

Tokenisation: Public blockchains enable the creation, issuance, and transfer of digital tokens representing assets such as real estate, securities,

and intellectual property rights, facilitating tokenization and fractional ownership.

Decentralised applications

(dApps): These blockchains host decentralised applications (dApps) across various industries, including gaming, social media, supply chain, and identity management, leveraging the benefits of decentralisation and censorship resistance.

Private vs public blockchain platforms

Private and public blockchain platforms offer distinct features and use cases tailored to different organisational needs and requirements. While private blockchains prioritise governance, privacy, and scalability for enterprise applications, public blockchains prioritise decentralisation, transparency, and censorship resistance for open and inclusive applications. By understanding the characteristics and use cases of each type of blockchain platform, organisations can leverage blockchain technology to drive innovation, efficiency, and transparency across diverse industries.

Table 2 lists the differences between public and private blockchains.

Table 2: The differences between public and private blockchains

Basis of difference	Public blockchain	Private blockchain
Access	Permissionless	Permissioned
Transparency	Fully transparent	Partially transparent
Decentralisation	Highly decentralised	Less decentralised
Security focus	External attacks, data integrity	Access control, internal fraud
Governance	On-chain governance	Off-chain governance
Scalability	Slower transaction processing	Faster transaction speeds
Cost	No fees (except for some transactions)	May require fees
Regulation	Less regulatory clarity	Stricter regulations
Use cases	Cryptocurrency, DeFi, supply chain, etc	Supply chain, trade finance, etc
Learning focus	Open participation, cryptoeconomics, etc	Permissioned access, governance, etc